Digital Twin Architecture

Concept

engine/digital_twin.py builds a live mirror of every resource.
Each resource has a ResourceState: cpu_usage, ram_usage, latency,
energy, security_risk, privacy_loss, trust_level, available.

Nodes

- LOCAL ..... the server itself (always present)
- MEC-001 ... MEC-0NN . every registered, non-revoked MEC device,
                        ONLINE via heartbeat (10 s cadence)
- CLOUD ..... the Google Drive / cloud node (always present)

Resource order: LOCAL, MEC-1.., CLOUD (config.RESOURCE_ORDER).

Synchronous twins

Heartbeat processing (process_heartbeat) updates the device twins in
real time; refresh_mec_status() flips offline devices after the
timeout so the scheduler never assigns to stale twins.